

# **The Texas City Refinery Disaster**

Homework No. 5 Che-396 - Senior Design 1  
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**Project Charter**

| Title                                                                                                                                 | Goal                                                                       | In Scope                                                    | Out of Scope                        |
|---------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------|
| The Texas City Refinery Disaster                                                                                                      | To choose the most beneficial case of action for all the parties involved. | Analysing all the information available about the disaster. | Laws, guide lines, regulations, etc |
| <b>Deliverables</b>                                                                                                                   |                                                                            |                                                             |                                     |
| Summary of disaster, problems in the refinery, ethical issues/improvements, case of actions, solution, and affects from the solution. |                                                                            |                                                             |                                     |
| <b>Team Members</b>                                                                                                                   |                                                                            |                                                             |                                     |
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**Design Plan**

| <b>Task ID</b> | <b>Task</b>                     | <b>Start Date</b> | <b>Durations</b> | <b>Finish Date</b> | <b>Members</b> |
|----------------|---------------------------------|-------------------|------------------|--------------------|----------------|
| 1.1            | Project Charter                 | 11/20/19          | 1 day            | 11/21/19           | all            |
| 1.2            | Design Plan                     | 11/20/19          | 1 day            | 11/21/19           | all            |
| 1.3            | Summary of Disaster             | 11/21/19          | 1 day            | 11/22/19           | Diego          |
| 1.4            | The problems                    | 11/22/19          | 2 days           | 11/24/19           | Diego          |
| 1.5            | Ethical Issues                  | 11/22/19          | 1 day            | 11/23/19           | Diego          |
| 2.1            | Case of Action 1                | 11/23/19          | 2 day            | 11/25/19           | Advith         |
| 2.2            | Case of Action 2                | 11/24/19          | 2 day            | 11/26/19           | Eric           |
| 3.1            | Why Case Action 1               | 11/25/19          | 2 days           | 11/27/19           | Jay            |
| 3.2            | Ethical Improvements            | 11/25/19          | 1 day            | 11/26/19           | Tariq          |
| 3.3            | Who is affected by the solution | 11/25/19          | 1 day            | 11/26/19           | Tariq          |

On March 23, 2005, a refinery located in, Texas City, Texas, was involved in one of the worst refinery disaster in the US. BP America was the owner of the refinery at the time of the disaster. The refinery is the third largest in the US and the second in Texas. Around 1:20 PM, a hydrocarbon vapor cloud was ignited and exploded killing 15 and injuring 180 others.

### ***1.1 - Summary of the disaster***

The night after the disaster, the isomerization unit (ISMO) was involved in a start-up process after a shut down for maintenance. [3] During the start-up, the raffinate splitter tower was being filled with flammable hydrocarbon liquids. Liquid levels had to be maintained at 6.5 feet according to procedure, but during start-ups, liquid levels reached 9 feet. This deviation was allowed because refinery operators had concerns regarding low liquid levels. [2] They believed that the low liquid levels could damage the furnace. During the night, two level control alarms failed and the liquid inside the tower filled to 13 feet. An hour before the end of the night shift, the operator of the, ISMO unit, left home early and left a vague entry in the log book about the start-up process. The morning operator, unaware of the conditions inside the tower, continued to pump liquid inside the tower. Also, the morning supervisor arrived an hour late to work and was never briefed about the conditions of the ISMO unit. During start-ups, the liquid of the tower had to be regulated with a level control valve but it was never opened. This is because the operator and others received conflicting instructions. The furnace was later turned on to heat up the feed of the raffinate tower and the morning supervisor left early on short notice. Another supervisor replaced the morning foreman but the replacement was not experienced with the ISMO unit. The morning operator was left, in charge of 3 process units, and left the ISMO unit unsupervised. As time passes by, the tower continued to fill up, causing a pressure control alarm to turn on. A manual pressure release valve was opened and the vapors were sent to a blowdown drum. Since no liquid was flowing out the tower, the operators opened the level control valve. The extremely hot liquid was sent to storage tanks, after it was used in a heat exchanger. In there, the hot liquid heated up the feed of the tower to a temperature of 141 F. As a consequence, the hot liquid of the tower began to swell and boil. Eventually, the liquid reached the top of the tower. The liquid started to overflow out of the tower into the vapor outlet. This caused, the pressure release valves to open and send the hot liquid to the blowdown drum. About 152,000 gallons of the liquid entered the blowdown drum and the liquid over filled into a process sewer. The process sewer was not enough and the level continued to rise. This was not noticed because the level control alarm, failed to turn on. At 1:20 PM the hot liquid and vapor erupted out of the drum. The amount of liquid and vapor released was equivalent to a tanker truck of hot gasoline. A vehicle

parked close, was left idling and the vapors of the drums entered the exhaust pipe and cause the engine to backfire. This ignited the vapor cloud and caused it to explode.

## ***1.2 - The problems***

The root cause of the issues that led to the accident stems from a lack of process safety management as well as the lack of a safety culture in the plant. Even preceding the accident, the Texas City refinery as a death trap with a reported 23 deaths in the preceding three decades. Additionally, there were numerous releases of hazardous material into the atmosphere with little or no changes made to day to day operations. Safety management focused solely on personnel safety and process safety was virtually nonexistent. Managers were only held responsible for the safety of individual worker's environment and were not concerned whether or not the process as a whole is safe. To add insult to injury, communication between operators, managers, contractors and even workers from different shifts was not enforced although mandated. Almost none of the people present were aware of the potentially dangerous isomar unit running the day of the incident. Additionally, there were contractors holding a meeting in a temporary trailer just 121 feet away from the blowback drum. This distance is unacceptable as people in trailers almost 500 feet away were injured There were no safe channels to communicate problems in without fear of losing one's job.

Another glaring issue directly caused by BP's business model, is all of the budget cuts and lack of investments in process safety. BP's goal in structuring the budget for American plants was to maximize profits, even if it threatens the integrity of the plant's safety. There were large budget cuts in 1999 and 2005 which may have influenced the severity of the accident. Operator and overall staffing was downsized. This meant less experienced personnel was present for each shift. Additionally, the downsizing greatly increased the workload of the plant operators, in turn creating stressful working conditions and fatigued operators. The operator who was working the day of the accident was on his 30 consecutive 12 hour shift. Also, operators would also veer from standard procedure in informal, unrecorded operation tasks. If the barebones crew wasn't bad enough, the equipment itself was another issue entirely. Although the equipment and instrumentation was scheduled to be maintained regularly, this schedule was not enforced and tasks could be marked as completed without actually being worked on. As well as not being maintained, the instrumentation was also calibrated incorrectly. The instrument reporting the level in the isomar unit were calibrated based on outdated data for a different liquid from a different process. This caused the inability of the operators to accurately read the level of liquid in the unit which allowed for a dangerous overfilling.

One of the more notable examples of outdated, poorly maintained equipment was the company's use of a blowback drum. A blowback drum is an outdated and now outlawed method of venting

hazardous hydrocarbons to the atmosphere. It was originally installed in the 1950's and was long overdue to be replaced as by company procedure. It remained in operation despite being unsafe. It is believed that the ignition source was a truck that was idling 25 feet away from the dangerous blowback drum. It is also believed that the two workers inhabiting the truck were unaware of operations.

### ***1.3 - Ethical issues***

Occupational Safety and Health Administration, or OSHA conducted an investigation into British Petroleum in right after the explosion that occurred in 2005. The purpose was to see if BP was operating correctly and if this explosion could have avoidable. Reports steamed that OSHA found 439 violations and penalties ranging up to \$31 million. To be more exact, hydrocarbons vapors came in contact with ignite source. This was due to the hydrocarbon atoms being heavier than the air molecules. There was a liquid overflow in one of the operating vessels, causing an overflow and overheating of the tower. It could be proven that the incidents that occurred in 2005 could have been avoided if the amount of violations were minimized. It was sad to research that in 2010 these similar rules were also violated in the oil spill incident in the Gulf of Mexico. In 2010 a BP platform exploded spilling 5 million barrels of oil and killing two workers. The aftermaths of this situation are still being experienced today. At the time of incident British Petroleum paid \$21 Million in fines and violations. Which is comparison to the oil spill in 2010, was much worse. Although BP did not violate as many violations as it did poses in 2005, the 2010 damage was much worse and therefore, that is why we see a \$10 Million increase in reprimands. In 2011 BP sold the refinery to make up for the cost.

### ***2.1 - Case of Action 1: Choosing to stop startup***

One route of action is to stop startup of the isomerization unit since proper protocol had not been followed. Inherently, startup of the isomerization unit is a dangerous process. As CSB stated in their investigation of the explosion “startup is an abnormal unit condition that requires significantly more manual control of a process, as well as critical thinking and decision-making that goes beyond normal unit operation.” [2] In the case of this specific startup, the plant operators deliberately skipped protocol in order to save time and were severely undermanned. This would be in line with the AiCHE code of ethics it is said that a professional should “hold paramount the safety, health and welfare of the public and protect the environment in performance of their professional duties.”

This action, if followed through by the other operators, would have likely saved the lives of the people who died that day. While operating the plant, each worker holds the lives of everyone nearby in their hands. This is without taking into consideration the ways in which dangerous plant operation indirectly affects people. A deadly explosion such as the one that happened affects the public opinion of BP, the stock price, and in the end the rest of the people working for BP. It would be taking a short term loss (cost of updating unit) for a long term gain.

Of course, there is a reason no one took this action. At the time BP had mandated 25% cost cut to fixed costs of production. Safety was simply not a priority to anyone, the goal was to produce as quickly as possible without any additional investment. It would be difficult for any one operator to take a stand against the entire culture of the plant. More likely, the stopping of procedure would have resulted in being reprimanded or fired. Realistically, BP needed changes made at a more executive level to emphasize safety culture and figure out a way to avoid cutting costs that impact safety.

### ***2.2 - Case Action 2: Name of Case Action***

The second course of action will be not to inform government authorities of this incident. By not reporting there will be benefits as well as consequences. One of the benefits will be the supervisor won't get fired and he/she will have income to support his /her family. The general idea in the plant was it hasn't gone wrong so far so why change it. These decisions do not follow any ethical guidelines as they are dangerous. The consequences as we know of neglecting important issues was the explosion of the plant. Life's were lost and many were injured from neglect.



### ***3.1 - Why Case Action 1***

The most important element in any industrial chemical process is the human element. That is why, regardless of which chemical is being produced, safety should be the number one priority. This is the main reason why stopping startup operations of the isomerization unit, until proper training is held and care is taken, would be the right decision for management. Disregarding all other variables, preservation of human lives come first. To minimize the possibility of another accident, additional, process safety minded staff must be hired. Also reworking management to mind process safety is necessary. While this is the right move, it would not have happened at the Texas City refinery because BP was not hospitable to operators or managers that would stop operations due to unwanted drops in production quantity. Although stopping operations would cause a short term loss, a safe, accident-free workplace would pay off in the long term in the form of happy productive workers and a better overall company image.

### ***3.2 - Ethic Improvements***

As times have passed, British Petroleum have changed their habits and have worked very closely with the Occupational Safety and Health Administration. BP has conducted follow up inspections and as a result had 270 violations after the incidents in 2010. As of now they only have 30 violations. BP did agree to pay \$50.6 Million to resolve the remaining citations. BP has paid over \$100 million in fines and violations, and had said to resolve all issues regarding their systems, but what stops mother nature from causing a disaster. As a result from the incidents that occurred in 2005, BP said they would remove all drums and vents used in the flammable process system. From reading about other oil rigs, they are all at risk for having an explosion or spilling oil. Oil rigs and vessels are easily targeted by the elements and can also be targeted by pirates. It has been reported that pirates have tried to take over oil rigs, although most are unsuccessful, these workers should never feel unsafe when coming to work. I believe that

### ***3.3 - Who is affected by the Solution***

US refineries, petroleum markets, and, most importantly, refinery workers will be affected by the solution. Safety should be a major priority for refineries because if accidents and disasters decrease, less money will be spent settling lawsuits. Cost cutting process safety and process personnel is not an economical solution. Many US refineries face the same problems as Texas City. If they spend money fixing and replacing old process equipment, the chances of an explosion will be reduced. This will allow the petroleum market to grow since the refineries will be able to operate at full capacity. In the case of Texas City, the refinery was forced to stop

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ISMO unit for 2 years, causing the refinery to lose billions of dollars. [3] What good does a refinery have if it cannot make money? Workers of refineries need a functional refinery to provide for their families. The loss of an employee's life is sad because his or her family will be left hopeless without them. Another reason is that most accidents can be avoided but unethical issues prevented workers lives to be spared. Training for new personal will also be beneficial to both the refinery and workers. A refinery operating under stress showed to be a disaster in Texas City. Refinery staff will be productive when working in tense free environments because without having pressure and lack of sleep, the employees will make sane decisions. [4] This will also allow employees to be well rested during work. The refinery workers will also have the opportunity to make more income for their families.

### References

- [1] <https://publicintegrity.org/business/workers-rights/worker-health-and-safety/renegade-refineries/bp-to-pay-13-million-for-safety-violations-at-texas-refinery/>
- [2] CSB. (2007). *Investigation Report: Refinery Explosion and Fire*. CSB.
- [3] “CSB Safety Video: Anatomy of a Disaster.” *YouTube*, YouTube, 2 June 2008, [www.youtube.com/watch?v=XuJtdQOU\\_Z4](http://www.youtube.com/watch?v=XuJtdQOU_Z4).
- [4] Mitchell, Jamie. “Sleep and Its Impact on the Modern-Day Workplace.” *Voluntary Protection Programs Participants' Association*, [www.vpppa.org/connect/blog/sleep-and-its-impact-on-the-modern-day-workplace#:~:targetText=One%20way%20of%20getting%20maximum,them%20to%20have%20enough%20sleep.&targetText=However%2C%20well%20Drested%20employees%20are,improves%20productivity%20in%20the%20process](http://www.vpppa.org/connect/blog/sleep-and-its-impact-on-the-modern-day-workplace#:~:targetText=One%20way%20of%20getting%20maximum,them%20to%20have%20enough%20sleep.&targetText=However%2C%20well%20Drested%20employees%20are,improves%20productivity%20in%20the%20process).